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# Unveiling Patterns: Adaptive Feature Selection and Sequential Clustering of Nanopore Signals

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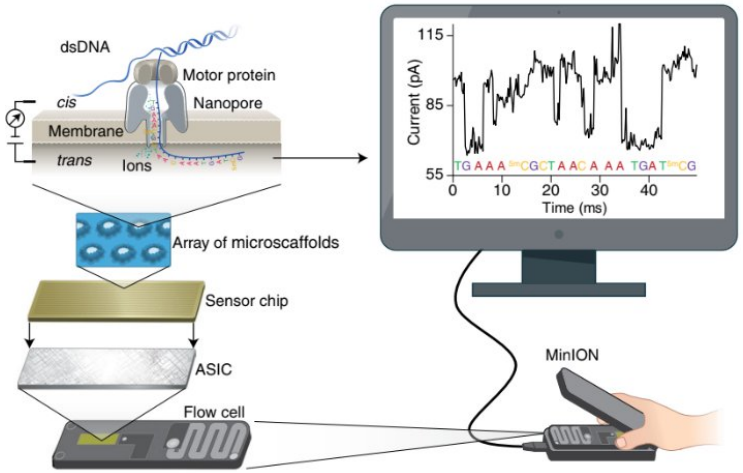
# Background

- Nanopore technology allows for the direct reading of nucleotide sequences as single molecules pass through nanoscale pores, eliminating the need for DNA amplification or synthesis
- The data structure reflects the sequential addition of analytes in a nanopore sequencing experiment
- Depending on the dataset, features that are used for clustering could vary, making it a challenge to identify the most relevant features



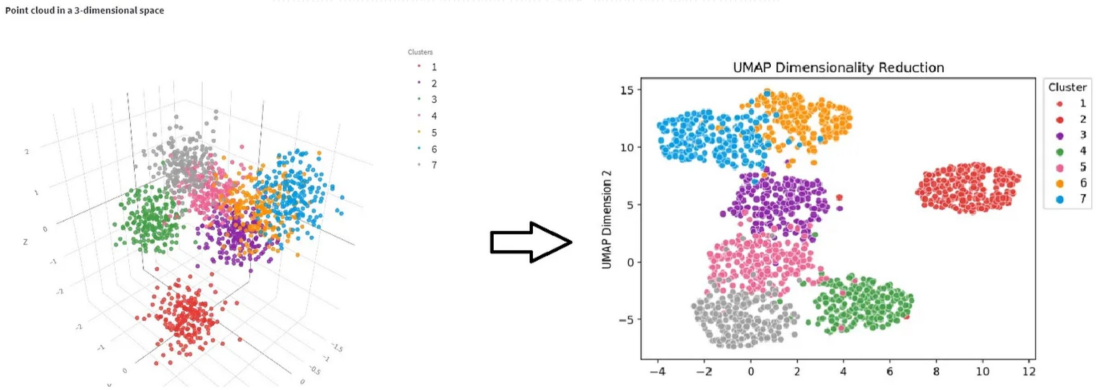

MinION flow cell [1]

# Goal

## Developing an adaptive feature-based clustering and pattern detection approach

- **The aim is to integrate feature selection directly into the clustering algorithm,** leveraging existing clustering algorithms like K-Means, Subspace Clustering, and more
- **We introduce analytes dynamically,** which can affect data distribution
  - We explore how subspace clustering can handle the sequential nature of data
- **We apply visualization techniques like UMAP**

**Secondary Objective:** DL methods like VAEs




UMAP visualization [2]

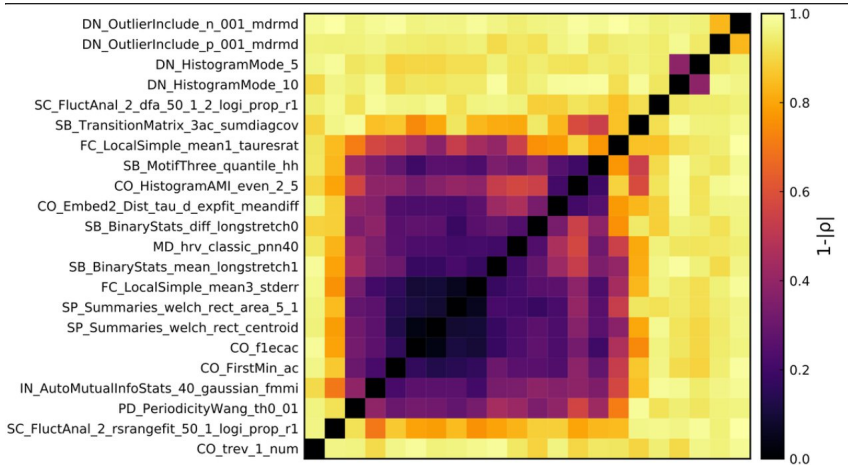
# Work Packages

- **Literature Review:**
  - We review literature on existing adaptive clustering approaches, to refine our approach
- **Exploratory Data Analysis**:
  - We examine the nanopore sequencing data available to us
  - Construct a feature pool from resources like the “Catch22” dataset and build upon it
  - Perform dataset specific analysis, and select features using unsupervised learning techniques
- **Integrated Feature Selection and Clustering**:
  - We integrate the feature selection into clustering so it is automatically performed
  - Use the feature pool from the previous work package

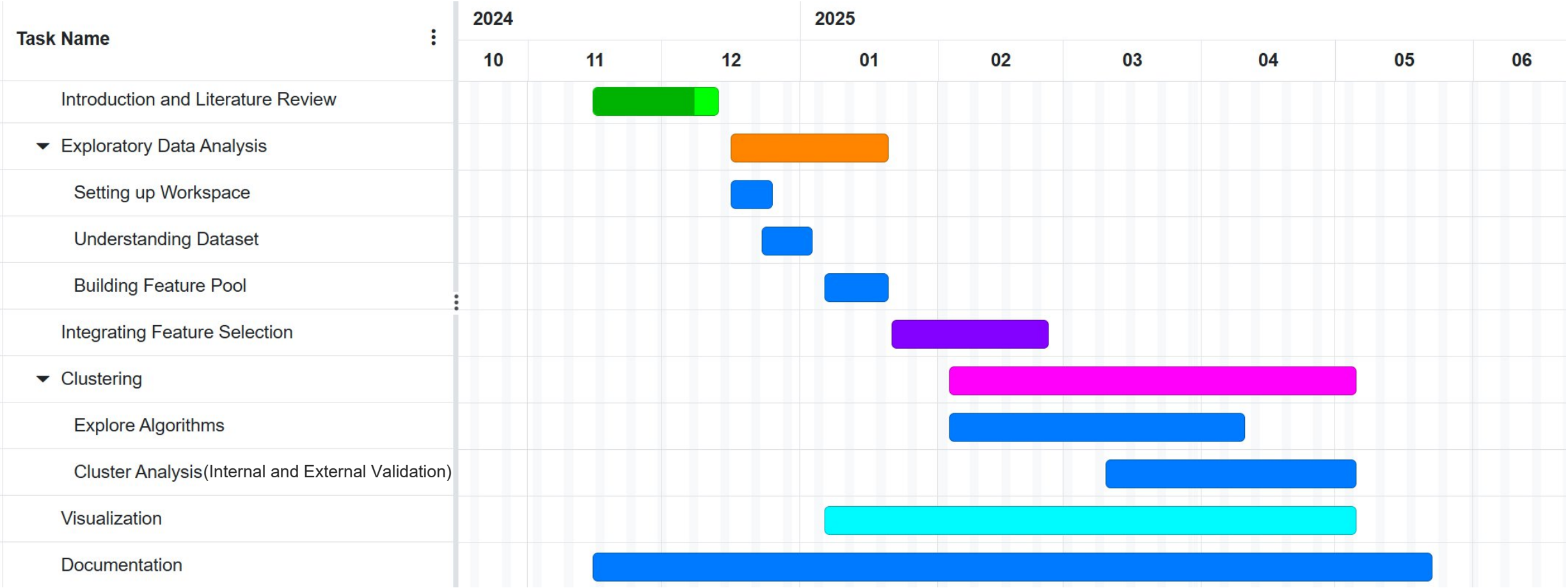
# Work Packages

- **Cluster Analysis and Interpretation:**
  - Use context from known data about the analytes
  - Conduct temporal analysis when analytes are added dynamically
  - Outlier detection
- **Visualization**
  - Use visualization techniques like UMAP, allowing us to assess cluster separation and understand how they evolve over time
  - Emphasize feature importance through methods like heatmaps

Catch-22 feature set [3]



# Timeline


Task Name
2024
2025
10
11
12
01
02
03
04
05
06
Introduction and Literature Review
Exploratory Data Analysis
Setting up Workspace
Understanding Dataset
Building Feature Pool
Integrating Feature Selection
Clustering
Explore Algorithms
Cluster Analysis(Internal and External Validation)
Visualization
Documentation

# References

[1] Wang, Yunhao, et al. "Nanopore sequencing technology, bioinformatics and applications." *Nature biotechnology* 39.11 (2021): 1348-1365.

[2] https://umap-learn.readthedocs.io/en/latest/plotting.html

[3] https://time-series-features.gitbook.io/catch22

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# Thank You